UF UNIVERSITY of FLORIDA

Master of Science in Computer Science

Concentrations in AI & Cybersecurity

Computer & Information Science & Engineering Department

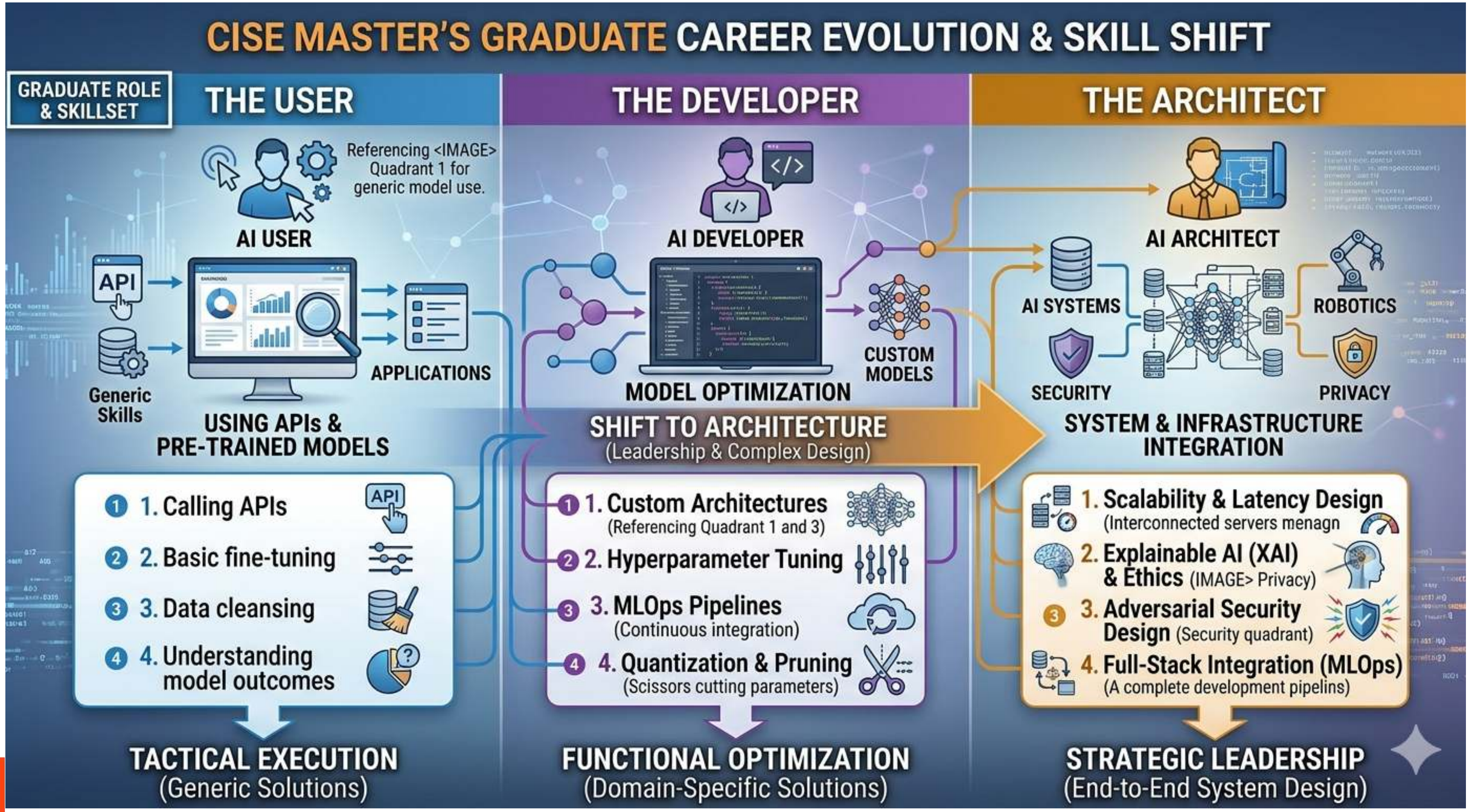
Director: Dr. Daisy Zhe Wang, PhD, Professor
Associate Director: Jennifer Reid, MBA

CISE@JAX Program Overview

- Master in Science in Computer Science with concentrations in AI & Cybersecurity.

- The program is 30 credits, full-time/part-time students
  - Time for completion: 1.5-2 years

- 2026 Fall: small cohort with students from working professionals and those freshly graduated from college. We expect the enrollment to grow significantly by fall 2027 and beyond when we can accept international students.

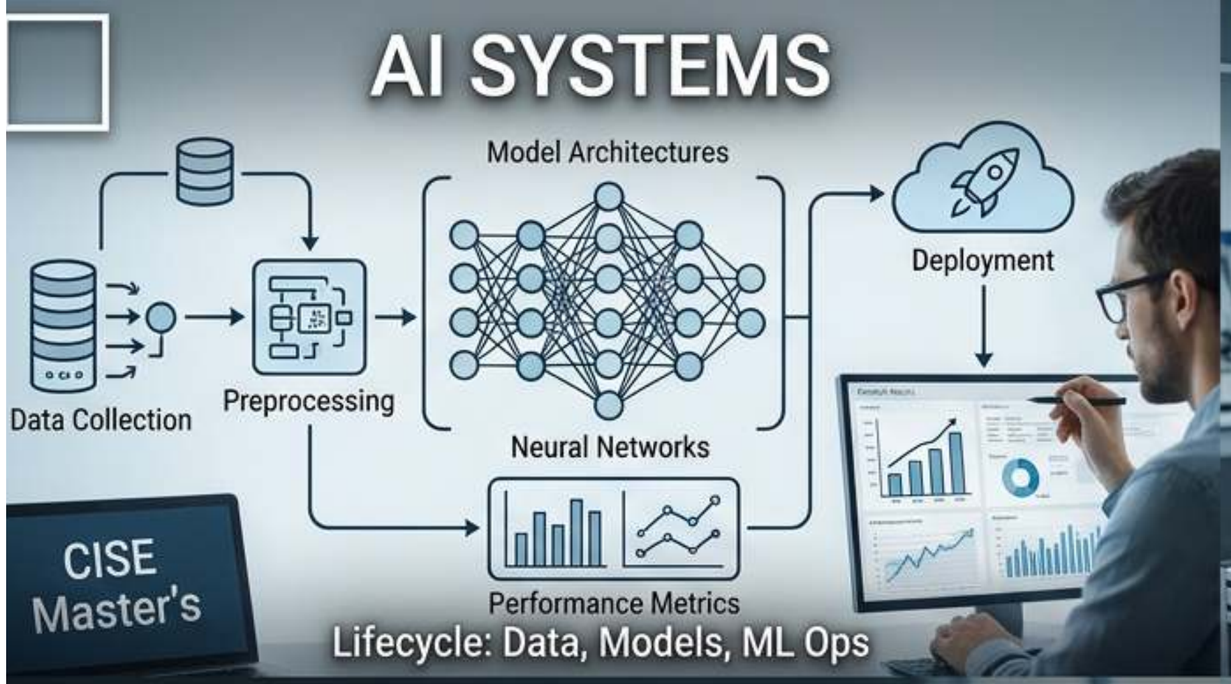

CISE MASTER'S GRADUATE CAREER EVOLUTION & SKILL SHIFT
GRADUATE ROLE & SKILLSET
THE USER
THE DEVELOPER
THE ARCHITECT
Referencing <IMAGE> Quadrant 1 for generic model use.
AI USER
API
Generic Skills
APPLICATIONS
USING APIs & PRE-TRAINED MODELS
AI DEVELOPER
MODEL OPTIMIZATION
CUSTOM MODELS
AI ARCHITECT
AI SYSTEMS
SECURITY
ROBOTICS
PRIVACY
SYSTEM & INFRASTRUCTURE INTEGRATION
SHIFT TO ARCHITECTURE
(Leadership & Complex Design)
1. Calling APIs
2. Basic fine-tuning
3. Data cleansing
4. Understanding model outcomes
1. Custom Architectures (Referencing Quadrant 1 and 3)
2. Hyperparameter Tuning
3. MLOps Pipelines (Continuous integration)
4. Quantization & Pruning (Scissors cutting parameters)
1. Scalability & Latency Design (Interconnected servers menagn
2. Explainable AI (XAI) & Ethics (IMAGE> Privacy)
3. Adversarial Security Design (Security quadrant)
4. Full-Stack Integration (MLOps) (A complete development pipelins)
TACTICAL EXECUTION
(Generic Solutions)
FUNCTIONAL OPTIMIZATION
(Domain-Specific Solutions)
STRATEGIC LEADERSHIP
(End-to-End System Design)

A Call to Action to Support CISE@JAX Programs

- The program emphasizes both theoretical foundations and practical applications. We are planning on experiential learning through both foundry/capstone projects as well as internships.

- CISE@JAX will provide research and development opportunities with world-class faculty and research institutions.

- **Engagement**: Career Development Programs, Experiential learning, Internships, Research Seminars/Symposiums/Expo, R&D Collaborations, EAB (External Advisory Board with industry and government partners)


AI SYSTEMS
Model Architectures
Deployment
Data Collection
Preprocessing
Neural Networks
Performance Metrics
CISE Master's
Lifecycle: Data, Models, ML Ops

ROBOTICS
CISE
Autonomous Systems, Control

ADVERSARIAL SECURITY
noise
Model visualization
Adversarial attacks
noise
Failure of faliure
Reinforced training
Attack Detection, Robust Training

AI PRIVACY
Federated Learning
local updates
Federated Learning
Federated Learning
local updates
data profile
data encryption
data encryption
Privacy-Preserving Tech, Compliance
data encryption
data vault
Federated Learning, Differential Privacy